

AI-POWERED CAD ANALYSIS REPORT

Generated: 12 Aug 2026

Geometry to Process to Cost · Fully AI-reasoned

100

Flange with 6-Hole Bolt Pattern and Central Boss

80 × 80 × 20 mm · 63.9 cm³ · AI 0.173 kg / Steel 0.502 kg

Manufacturability: 100/100 · Confidence: Low

§1 — Geometry & Part Summary

Bounding Box	80.0 × 80.0 × 20.0 mm	Surface Area	148.4 cm ²
Volume	63.94 cm ³	Weight (AI)	0.173 kg
Weight (Steel)	0.502 kg	Weight (Plastic)	0.067 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Bolt holes	6	High	6x through-holes, r=3mm, on bolt-circle pattern
Central boss	1	High	Cylindrical boss, r=40mm, likely spigot/register feature
Secondary boss/step	1	Medium	Cylindrical shoulder, r=15mm
Zero-draft walls	8	High	8 faces with 90° draft angle — problematic for die/pattern draw
Thick non-uniform section	1	High	Wall 11.95 - 19.95 mm, Å = 4 mm — hot-spot/shrinkage risk

§3 — Material Analysis

AI-suggested

Primary Material:

Ductile (SG) Iron GJS500 (55% confidence)

Thick, non-uniform sections (12-20mm), a 6-bolt flange with central register boss, and zero-draft cylindrical faces are classic sand/gravity-cast ductile-iron hub/flange geometry rather than thin-wall HPDC aluminium (which needs 2-4mm uniform walls). The geometry pack itself supplies a ferrous-specific sand cycle time (0.168hr) alongside the generic one, signalling a ferrous material intent. Per guidance, when uncertain between aluminium and a dense structural metal on a thick, bolt-flanged hub-like part, defaulting to a ferrous casting alloy (ductile iron, cheaper and more castable than steel at this section thickness) is the safer assumption; wrought/forged steel was not selected because Stage-1 classified this firmly as a casting process (0.75 conf), and cast steel is rarely used for a part this size/volume when ductile iron performs equivalently at lower cost.

Alternatives:

Aluminium LM25 (cast) (25%) · Grey Iron GJL350 (12%) · Austempered Ductile Iron (8%)

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Sand casting (high-pressure automated line) + machining	casting	70%	0.1680	Thick, non-uniform ferrous section with bolt flange geometry suits automated green-sand or shell moulding at high volume; matches supplied ferrous sand cycle time.
Gravity die casting (if aluminium confirmed)	casting	20%	0.1570	If material is later confirmed aluminium, thick sections (12-20mm) and moderate volume still suit permanent-mould gravity casting rather than thin-wall HPDC.
CNC finish machining of flange faces/holes/bore	machining	90%	0.1320	Cast blank requires facing, boss bore finishing and hole sizing per bottom-up CNC estimate.

§5 — Manufacturability Risks (Score: 100/100)

Severity	Feature / Area	Description	Recommended Action
High	Zero-draft faces (8)	8 faces measured at exactly 90° draft — will gall/stick in a permanent mould or require pattern modification for sand tooling	Add 1-3° draft to all vertical cylindrical/planar faces along the draw direction
Medium	Non-uniform wall (A = 4 m m)	Wall ranges 11.95-19.95mm creating hot spots prone to shrinkage porosity, especially under the 40mm boss	Add core-out/coring under the boss or blend fillets to even out cooling gradients; place risers over heavy sections
Low	6x small holes (r=3mm)	Small cored holes may require thin core pins prone to shift/breakage in sand moulding	Cast solid and drill post-machining, or use robust core prints with adequate support

§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£0.00	£0.00	£0.00

Net Weight	0.457 kg	Material	mat-gjs500
Recommended Process	casting	Cycle Time	0.1320 hr/part
Setup Time	0.500 hr	Operations	2 ops
Operations Detail	Mill flange faces (mach-vmc3, 0.0313 hr) Drill/bore 6 holes + boss ID (mach-drill, 0.0489 hr)		

Process-Specific Parameters

Casting Subtype	sand
Die/Mould Cost	£2,500
Die Life	8,000 shots
Cavities	4
Yield	65.0%
Flash Weight	0.000 kg
Yield	0.0%
Die Cost	£0
Strokes	0
Cavities	0
Wall Thickness	0 mm
Mould Cost	£0
Mould Life	0 shots
Projected Area	0.0 cm²

§8 — AI Analysis Explanation